DTU

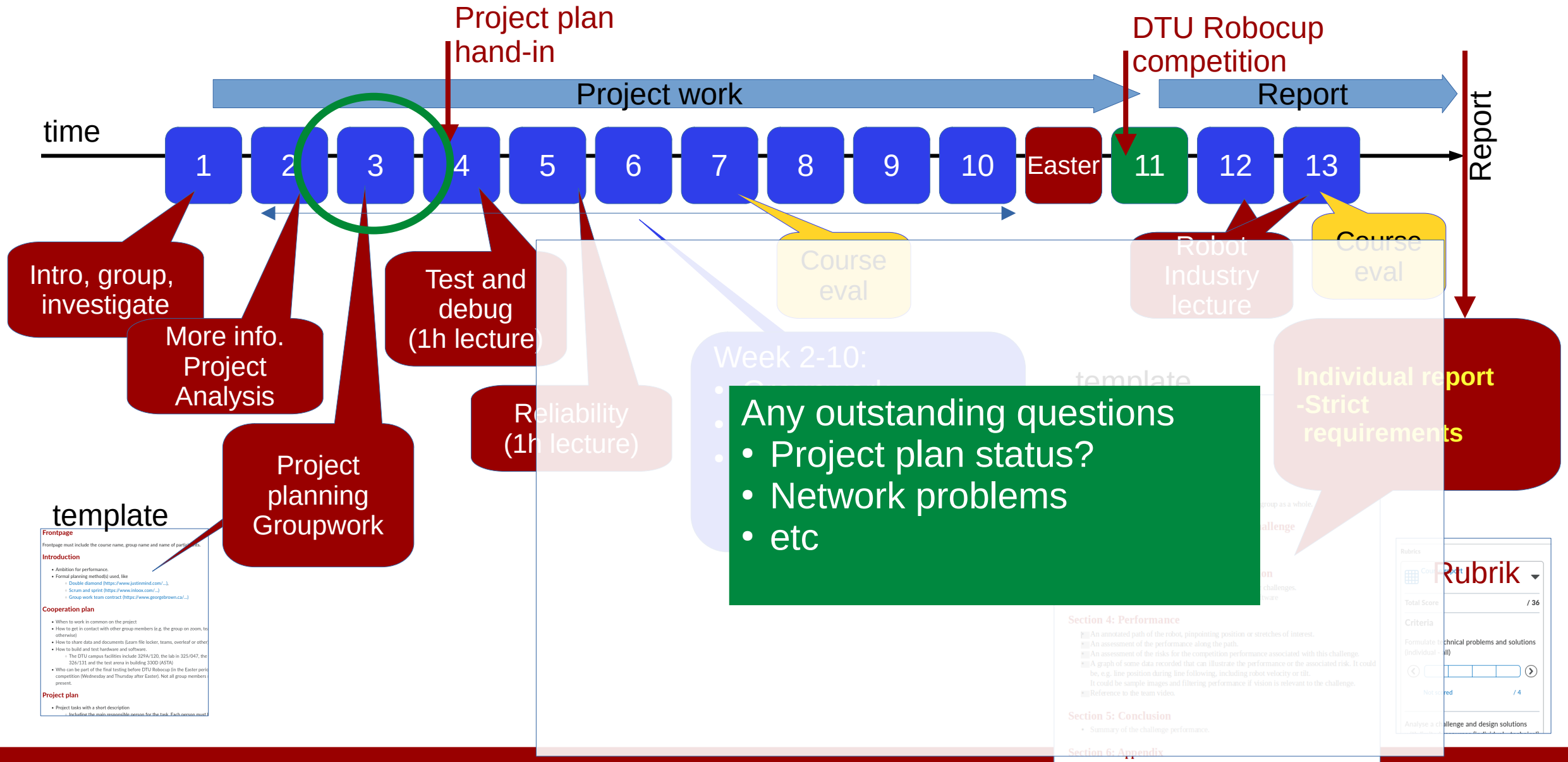
34755 Building dependable robots

- Today
  - Course status
    - Messages
    - Questions before we start
  - Introduction to project planning and execution
    - Project plan
    - Double diamond and Scrum
  - Group work
  - Solve the quiz of the day

J. Christian Andersen
PCAS
DTU Electro

34755 – Building dependable robot systems

2025



Any outstanding questions

- Project plan status?
- Network problems
- etc

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Project plan

Deliverable - *next* ***Thursday 26 Feb*** *at 23:59*

- Create a project plan *for the remaining period*.

Content

Front page

- Introduction
- Cooperation plan
  - Cooperation tools and agreements
- Project Plan for the group
  - Tasks to do
  - Resource allocation (task to person(s))
  - Management responsibilities
- Individual challenge for the final report
  - Which single challenge or subject.
  - Can be tentative.
- Evaluation
  - There is a Rubric for the plan

What counts in the end:

- Project plan: 15% (group)
- Final report: 40% (individual)
- Half way competition 7% (group)
- Final competition 18% (group)
- Solved quizzes: 10% (individual)
- Group video: 10% (group)

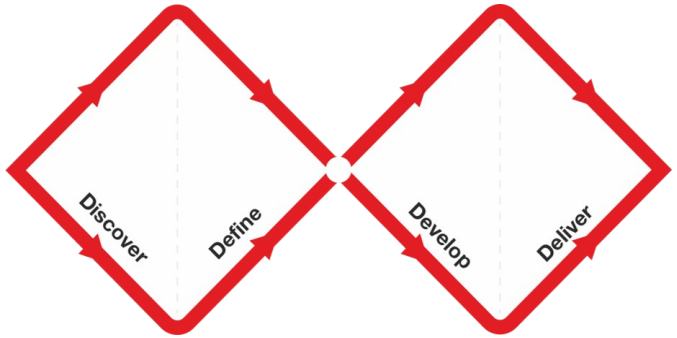
Template

You could use the template for the final report (Latex for overleaf)

5-10 pages
+ front page

Project plan - Introduction

Double Diamond

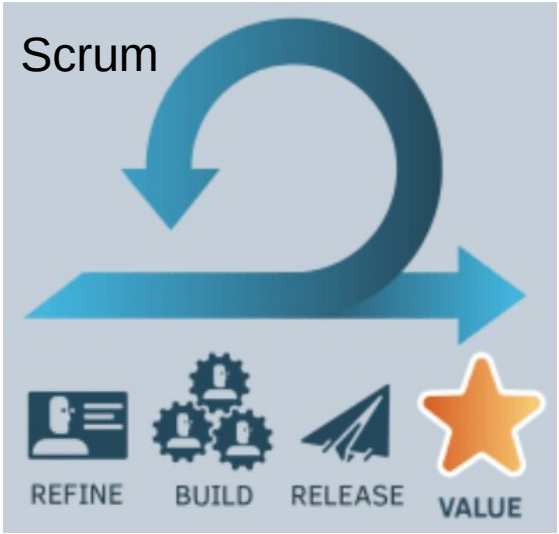


Front page
- **Introduction**
- Cooperation plan
- Project Plan
- Individual challenge

Introduction

- Group introduction and project vision for the group.
- Comment on the project planning and execution method:
  - How to use Double Diamond (e.g. see Double diamond design report),
  - How to use Scrum and Sprint (https://www.scrumalliance.org/about-scrum)

Scrum



I will get back to Double Diamond and Scrum later today

1-1½ pages

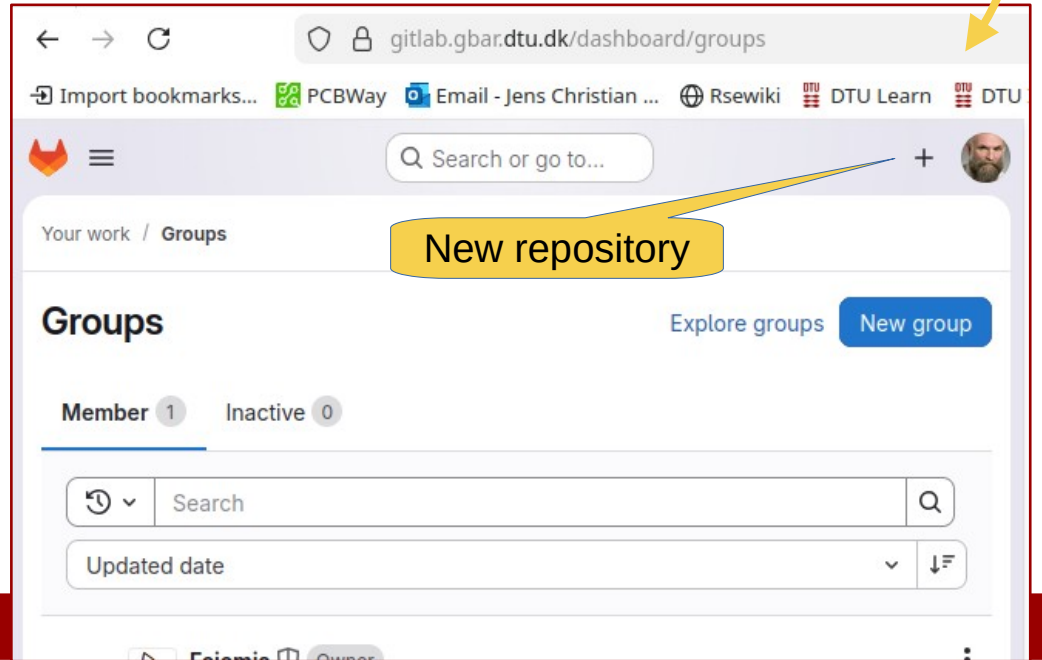
Project plan - Cooperation plan

Cooperation plan

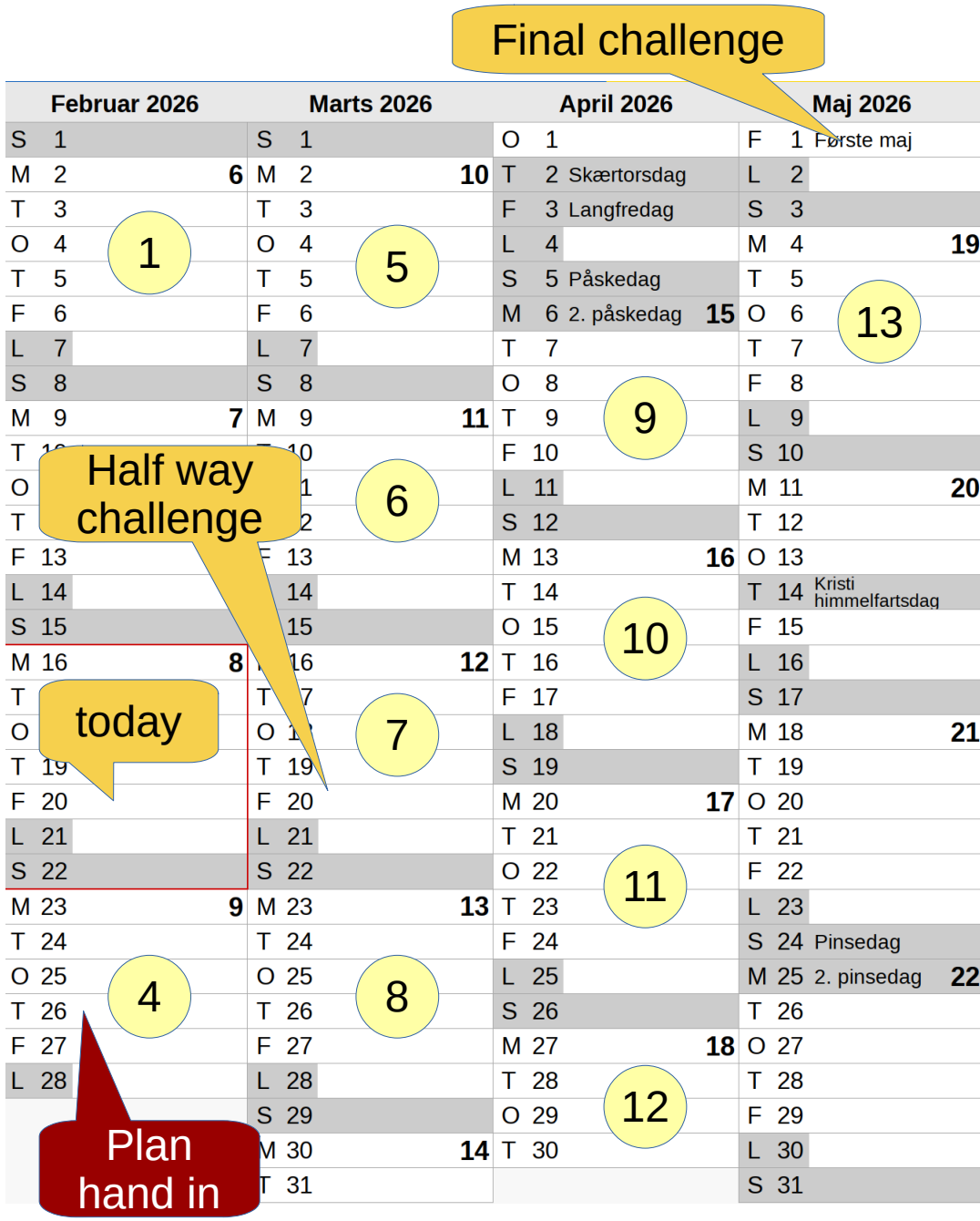
Elements from a group contract like:

(e.g. from https://www.georgebrown.ca/...)

- When to work
- How to get in contact
- Which repository (e.g. https://gitlab.gbar.dtu.dk/)
- Where to build hardware and software.



1-2 pages



Project plan

Front page
- Introduction
- Cooperation plan
- **Project Plan**
- Individual challenge

Project Plan
- Tasks to do and who to do it, like:
  - Vision
  - Use of ArUco
  - Driving on tape-line
  - Designing actuator(s) (ball and drone capture)
  - Calibrate sensors
  - Recovery plans
  - Project video
  - Competition challenges
  - Task priority
  - …
- Management and special roles, like:
  - CPO, Product coordinator (product, moderator)
  - CTO, Test manager (Robot available for all)
  - CLO, Software Library officer (GIT)
  - CCO, Communication Officer, internal and external
  - CMO, Multimedia Officer (e.g. Video)
  - ...

2-4 pages

Plan hand in

Final challenge

Half way challenge

today

Februar 2026		Marts 2026		April 2026		Maj 2026	
S 1		S 1		O 1		F 1	Første maj
M 2	6	M 2	10	T 2	Skærtorsdag	L 2	
T 3	1	T 3	5	F 3	Langfredag	S 3	
O 4		O 4		L 4		M 4	19
T 5		T 5		S 5	Påskedag	T 5	
F 6		F 6		M 6	2. påskedag 15	O 6	13
L 7		L 7		T 7		T 7	
S 8		S 8		O 8	9	F 8	
M 9	7	M 9	11	T 9		L 9	
T 10		T 10		F 10		S 10	
O 11		11	6	L 11		M 11	20
T 12		12		S 12		T 12	
F 13		F 13		M 13	16	O 13	
L 14		14		T 14		T 14	Kristi himmelfartsdag
S 15		15		O 15	10	F 15	
M 16	8	16	12	T 16		L 16	
T 17		T 17		F 17		S 17	
O 18		O 18	7	L 18		M 18	21
T 19		T 19		S 19		T 19	
F 20		F 20		M 20	17	O 20	
L 21		L 21		T 21		T 21	
S 22		S 22		O 22	11	F 22	
M 23	9	M 23	13	T 23		L 23	
T 24		T 24		F 24		S 24	Pinsedag
O 25	4	O 25	8	L 25		M 25	2. pinsedag 22
T 26		T 26		S 26		T 26	
F 27		F 27		M 27	18	O 27	
L 28		L 28		T 28	12	T 28	
		S 29		O 29		F 29	
		M 30	14	T 30		L 30	
		T 31				S 31	

Project plan - Challenge allocation

Front page
- Introduction
- Cooperation plan
- Project Plan
- **Individual challenge**

Challenge allocation

- List group members and allocated individual report challenge.
  - Each person write a report on one of the challenges/subject.
  - **Separate from** other group members.
  - Intended for after the competition.
  - Data collection before or at the competition preferred.

½ page

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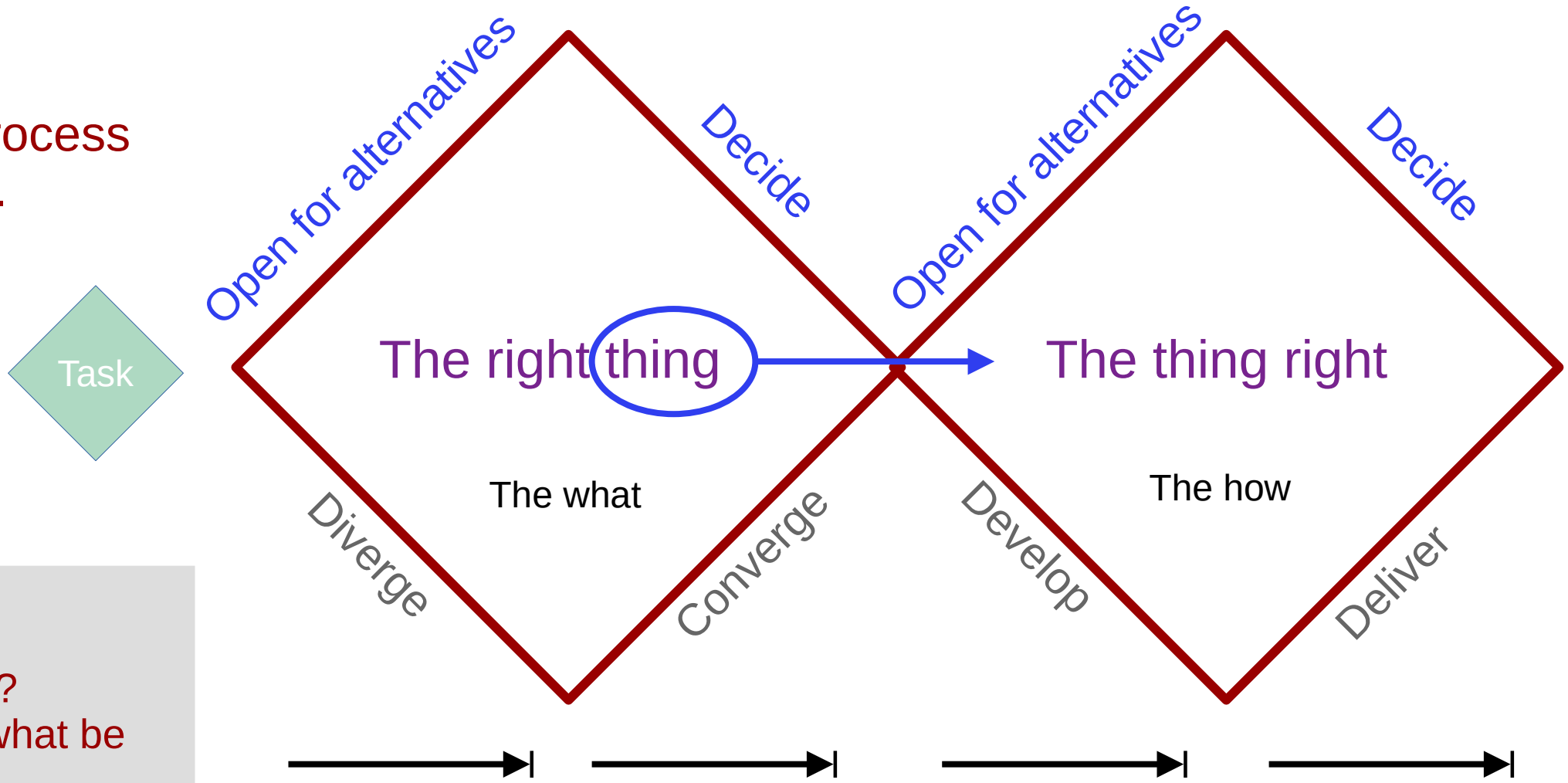
From https://www.designcouncil.org.uk/our-resources/framework-for-innovation/

Double Diamond

A design process description.



e.g.:
Project plan:
- Who do what?
- How should what be done?

Someone should say: Time to go to the next phase.
(but redo poor decisions take more time)

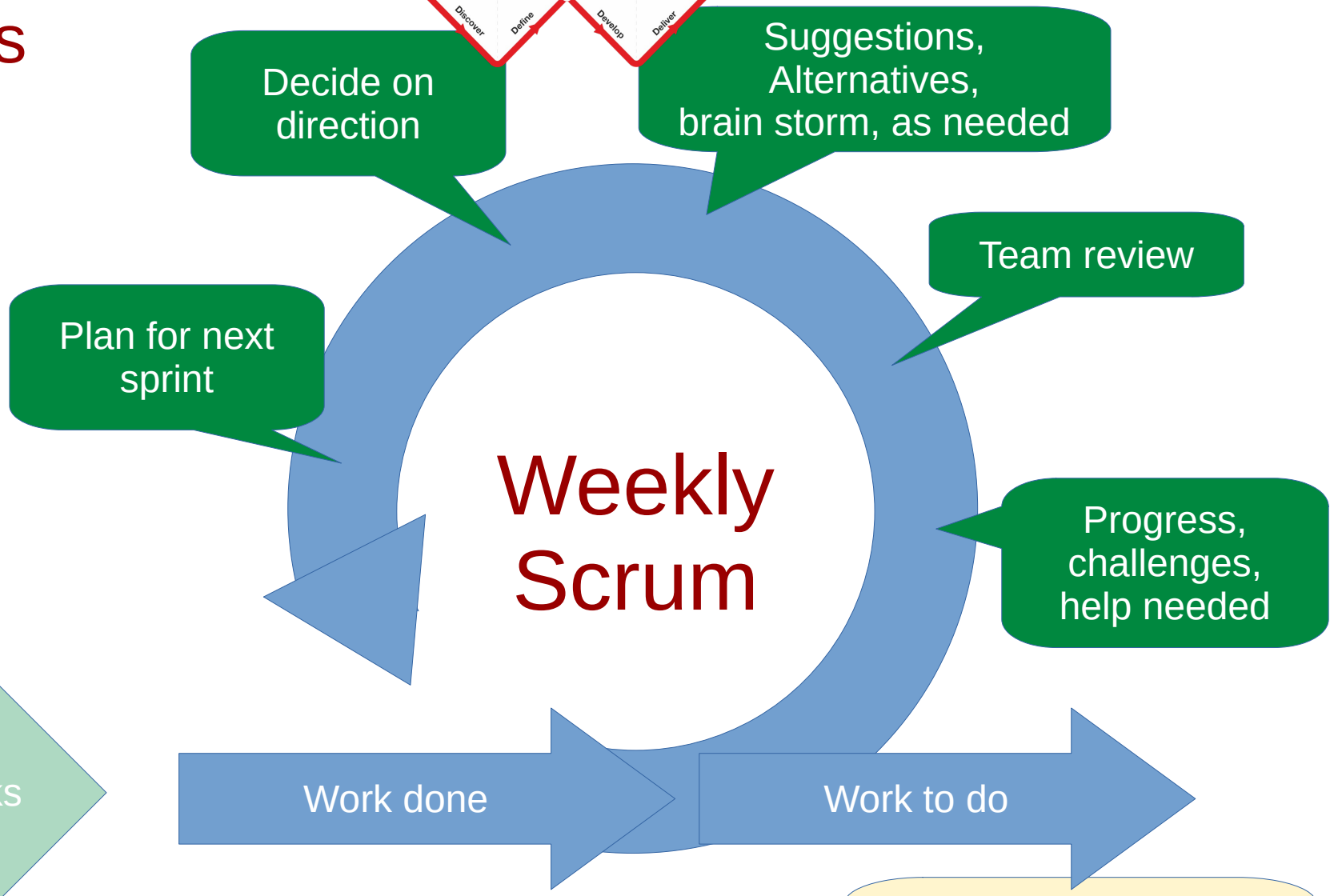
Scrum and Sprints

A development framework.

Scrum suggestion:

- Full group meet every week
- Everyone present progress
- Group review and suggest
- Everyone present plan to next sprint

From https://www.scrumalliance.org/about-scrum



Group sprint: less than one hour (mostly)

Note: process assessment is needed in the final report

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